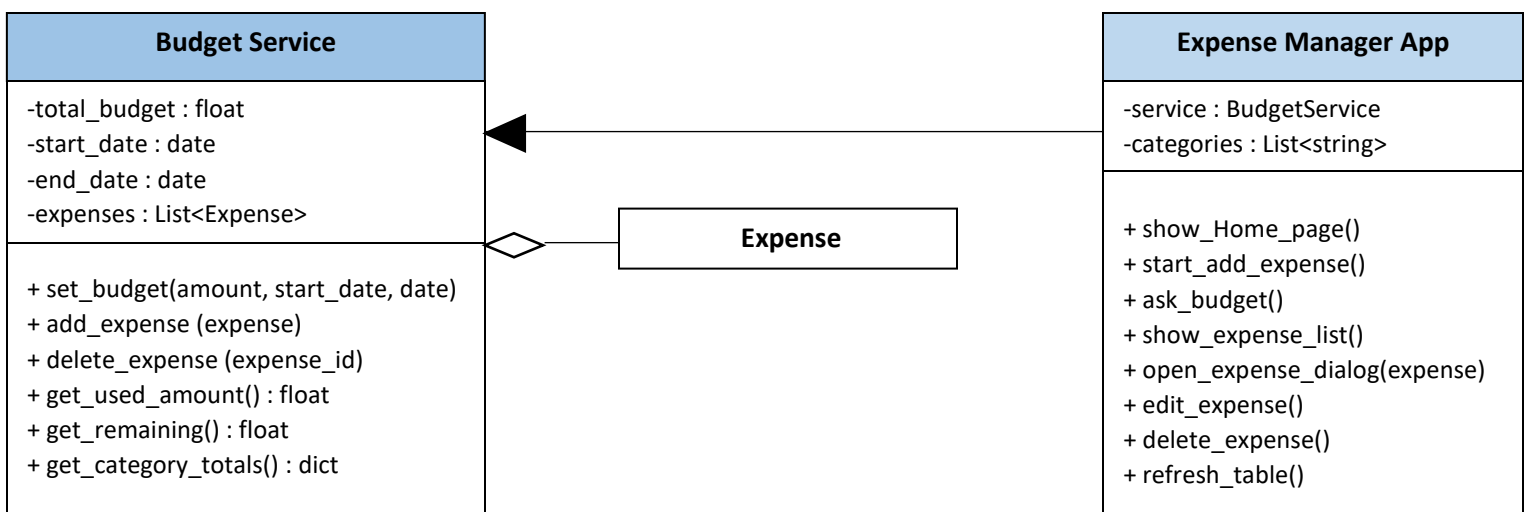
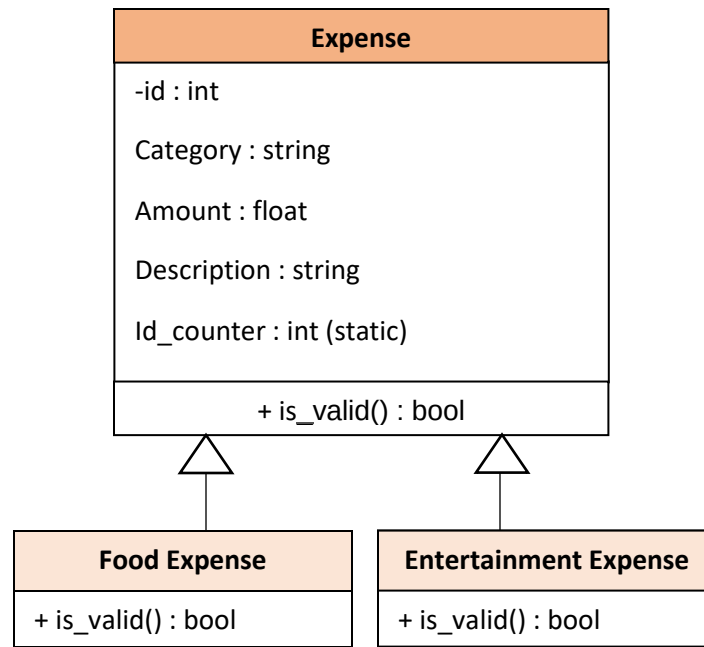


UML CLASS DIAGRAM



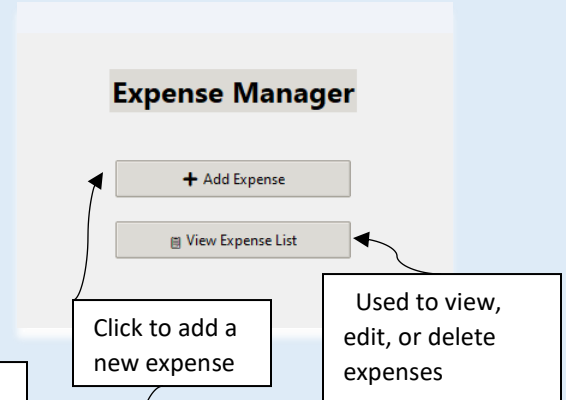
USER MANUAL

The "Expense Manager" is a desktop application that will help users manage their expenses according to a specific budget. The application will enable users to create a budget for a specified time period, add expenses for specific categories, view all expenses in a list format, and monitor the remaining budget in real time.

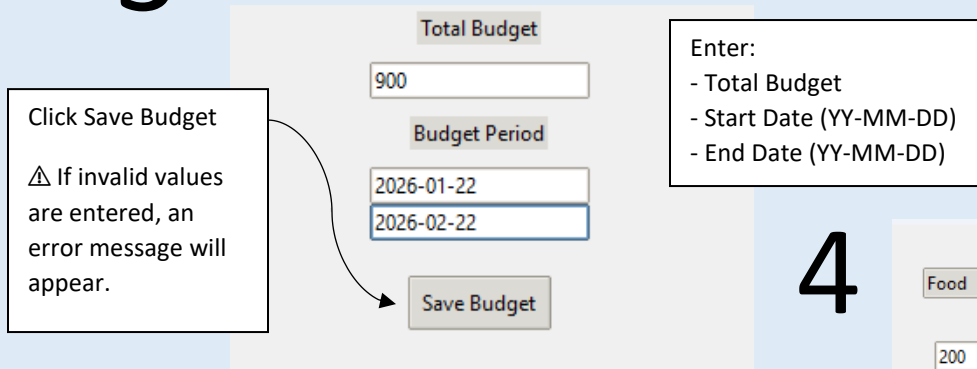
1 SYSTEM REQUIREMENTS

- Operating System: Windows / macOS / Linux
- Python 3.x installed
- Required libraries:
- tkinter (GUI)
- datetime (data handling)

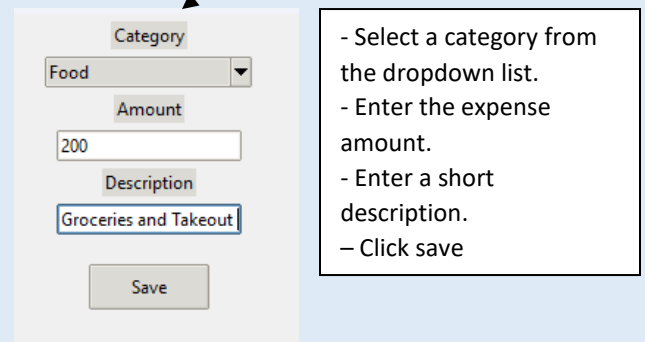
2 Run the Python file containing the application code and the home window titled "Expense Manager" will open.



3 When adding an expense for the first time, the user is prompted to set a budget.

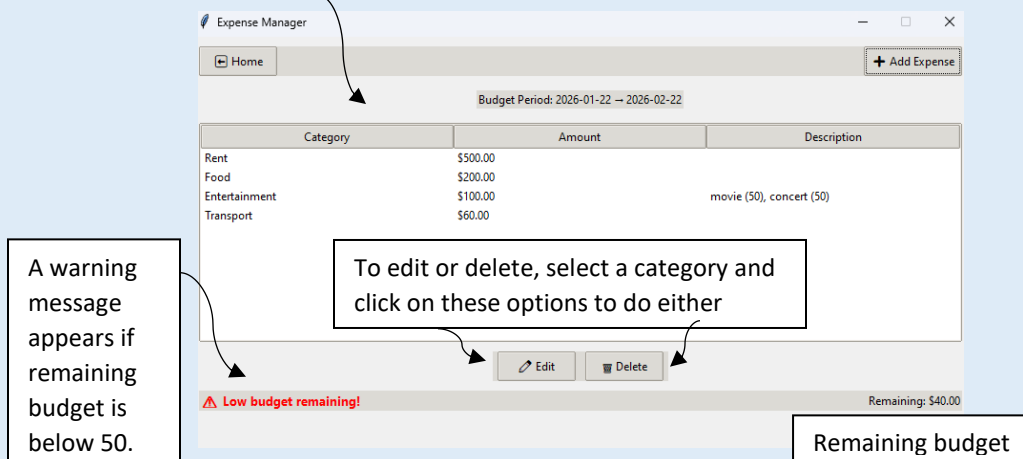


4



5 Viewing the Expense

A table will display all added expenses with: Category, Amount, Description



ERROR HANDLING-

The application shows error messages when:

- Invalid budget values are entered
- Expense amount exceeds category limits
- No expense is selected for edit or delete

UNIT TEST REPORT

Unit tests were implemented using Python's **unittest** framework. The tests verify expense validation logic and polymorphic behavior across different expense types. All unit tests were executed successfully, as shown below.

```
C:\Users\Faisal Hashmi>cd "C:\Users\Faisal Hashmi\Desktop\Uni Assignments\Programming II"
C:\Users\Faisal Hashmi\Desktop\Uni Assignments\Programming II>python test_expense_manager.py
....
-----
Ran 4 tests in 0.001s

OK
```